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Method for Producing Conventionally Hot-Rolled Profiled Strip Products

Abstract

The invention relates to a method for producing hot-rolled hot strip products in which a steel alloy is melted; the melted steel alloy is cast into slab ingots and after being heated to a temperature above $AC_{sub.3}$, the slab ingots are hot rolled until they reach a desired degree of deformation and a desired strip thickness; the rolling is performed above the recrystallization temperature of the alloy; after the rolling, the strip is cooled to room temperature and for hardening purposes, is briefly heated to a temperature $>Ac_3$ and cooled again, characterized in that the heating takes place with a temperature increase of more than 5 K/s, more than 10 K/s, more than 50 K/s, or more than 100 K/s and is kept at a desired target temperature for a period of 0.5 to 60 s before cooling to yield improved mechanical properties. and then a cooling takes place;

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Background/Summary

RELATED APPLICATIONS [0001] This patent application is a continuation of U.S. application Ser. No. 17/413,435, filed on Jan. 14, 2022, which in turn is a 35 U.S.C. § 371 National Stage entry based on and claiming priority to International Application PCT/EP2019/086062, filed on Dec. 18, 2019, which in turn claims priority based on German Application DE 10 2018 132 860.5, filed on Dec. 19, 2018, the disclosures of which are incorporated by reference in their entireties.

FIELD OF THE INVENTION

[0002] The invention relates to a method for producing conventionally hot-rolled and profiled or formed hot strip products as described and claimed herein.

BACKGROUND OF THE INVENTION

[0003] The term “hot strip” refers to a hot-rolled steel strip which is produced in that first, a steel melt of a desired alloy along with inevitable impurities that are intrinsic to the steel melting process is melted, generally in a converter. The melt is usually then fed into a metallurgical ladle in which a melt metallurgical processing, in particular an alloy adjustment, takes place. A wide variety of oxidizing processes are also carried out in the converter in order to reduce the quantity of unwanted by-elements.

[0004] Then the steel is poured from the ladle usually via a tundish into a continuous casting machine in which the liquid steel is cast into a theoretically endless strip of slab ingots. In the continuous casting machine, the solidified steel strip is cut into so-called slab ingots, which are slab-shaped, with a thickness of several decimeters, a width of for example 1.5 m, and a length of for example 6 to 12 m.

[0005] Such slab ingots can then be further processed in rolling trains.

[0006] To produce a hot strip, such slab ingots are first preheated to the rolling temperature in a reheating furnace and then travel into the so-called hot (wide) strip mill.

[0007] The hot strip mill consists of a series of rolling stands; first, there is a so-called reversing roughing mill in which the slab ingot is rough-rolled. The still very hot, brightly glowing steel strip is then fed into the actual rolling stands and passes through these rolling stands, which give the strip a target thickness and target width.

[0008] Hot strips of this kind, after they have been coiled, can either undergo further processing immediately or they can be processed into steel sheet by means of a cold-rolling train.

[0009] Hot strip, however, is not only produced strictly for processing into steel sheet, but also

represents its own steel specialty product, which can undergo immediate processing with modifications.

[0010] In the conventionally produced hot strip, the greatest part of the forming takes place above the recrystallization stop temperature, as a result of which the austenite develops a globular grain form at the end of the rolling process, as shown in FIG. 1.

[0011] As shown in FIGS. 6a and 6b, in the conventionally hot-rolling with a plurality of roll passes, rolling is performed at a temperature above the recrystallization temperature and then a cooling is performed or quenching is performed using the so-called direct quench method. The resulting possible structures are globular austenite above the recrystallization temperature, see FIG. 3, which after the cooling, in particular after direct quenching (quenching from the rolling heat) transforms into martensite or annealed martensite. In a subsequent hardening step, the globular austenite once again forms martensite, which then after the annealing, exhibits an annealed martensite structure. If the hardening step is achieved (FIG. 6b) by means of a quenching directly from the rolling heat, all that is needed is an annealing in order to thus produce an annealed martensite structure.

[0012] These standard treatment routes can be used to adjust certain properties such as toughness and strength of such a material.

[0013] WO2017/016582 A1 has disclosed a high-strength steel with a high minimum yield strength and a method for producing a steel of this kind. This steel has a composition that comprises the following: [0014] (a) carbon: 0.23 to 0.25 wt % [0015] (b) silicon: 0.15 to 0.35 wt % [0016] (c) manganese: 0.85 to 1.00 wt % [0017] (d) aluminum: 0.07 to 0.10 wt % [0018] (e) chromium: 0.65 to 0.75 wt % [0019] (f) niobium: 0.02 to 0.03 wt % [0020] (g) molybdenum: 0.55 to 0.65 wt % [0021] (h) vanadium: 0.035 to 0.05 wt %; [0022] (i) nickel: 1.10 to 1.30 wt %; [0023] (j) boron: 0.0020 to 0.0035 wt %; [0024] (k) calcium: 0.0007 to 0.0030 wt %; and the steel possibly contains other elements, with the maximum concentration of the other elements being: [0025] (l) phosphorus: <0.012 wt % and/or [0026] (m) sulfur: <0.003 wt % and/or [0027] (n) copper: <0.10 wt % and/or [0028] (o) nitrogen: <0.006 wt % and/or [0029] (p) titanium: <0.008 wt % and/or [0030] (q) tin: <0.03 wt % and/or [0031] (r) hydrogen: <2.00 ppm and/or (s) [0032] (s) arsenic: <0.01 wt % and/or [0033] (t) cobalt: <0.01 wt %; the rest comprising iron and inevitable impurities; and [0034] (i) the carbon equivalent P_{cm} can be calculated as follows

[00001]

$$P_{cm} = [C] + [Si] / 30 + [Mn] / 20 + [Cu] / 20 + [Ni] / 60 + [Cr] / 20 + [Mo] / 15 + [V] / 10 + 5[B];$$
where [C], [Si], [Mn], [Cu], [Ni], [Cr], [Mo], [V], and [B] are the mass fractions of the respective elements in the high-strength steel in wt % and where for P_{cm} the following relation applies: 0.38 wt % < P_{cm} < 0.44 wt %; and/or [0035] (ii) the carbon equivalent C_{eq} can be calculated as follows

[00002]
$$C_{eq} = [C] + [Si] / 24 + [Mn] / 6 + [Ni] / 40 + [Cr] / 5 + [Mo] / 4 + [V] / 14;$$

where [C], [Si], [Mn], [Ni], [Cr], [Mo] and [V] are the mass fractions of the respective elements in the high-strength steel in wt % and where for C_{eq}, the following relation applies:

0.675 < C_{eq} < 0.78 wt %; and/or [0036] (iii) the carbon equivalent CET can be calculated as follows

[00003]
$$CET = [C] + ([Mn] + [Mo]) / 10 + ([Cr] + [Cu]) / 20 + [Ni] / 40$$

where [C], [Mn], [Cr], [Mo], [Cu] and [Ni] are the mass fractions of the respective elements in the high-strength steel in wt % and where for CET, the following relation applies: 0.43 wt % < CET < 0.49 wt %.

[0037] During production, the hydrogen content is reduced by means of a vacuum treatment of the steel melt after which the steel melt is cast into a slab ingot. The slab ingot is then heated to a temperature in the range from 1100° C. to 1250° C., descaled, and then hot rolled into a flat steel product. The product is then coiled; the coiling temperature is at least 800° C.; during the hot rolling of the slab ingot into a flat steel product, the initial rolling temperature is in the range from 1050° C. to 1250° C. and the final rolling temperature is ≥ 880° C.; and for the P_{cm}, the following

relation applies: $0.38 \text{ wt \%} < \text{Pcm} \leq 0.44 \text{ wt \%}$. After the hot rolling, the flat steel product preferably is subjected to a hardening treatment; the hardening treatment is performed at a temperature of at least 40 Kelvin above the Ac_3 temperature of the steel alloy and the flat steel product is then quickly quenched so that the cooling speed is at least 25 K/s, is at a temperature that lies below 200°C . The minimum austenitization temperature of the flat steel product according to WO2017/016582 A1 for the uniform austenitization is $\geq 860^\circ \text{C}$. Lower austenitization temperatures $< 860^\circ$ in combination with the balanced chemical composition of this steel alloy result in an unwanted partial austenitization. Preferably, the austenitization temperature should be $\leq 920^\circ \text{C}$; higher temperatures promote austenite grain growth, which results in a reduction in the mechanical/technological properties. The optimal austenitization temperature should be 880°C . [0038] EP 2 267 177 A1 discloses a high-strength steel plate, which is used as a structural element in industrial machines and which on the one hand, should have an outstanding resistance to a delayed fracture and on the other, should have a good welding behavior. This steel plate has a minimum yield strength of 1300 MPa or greater and a tensile strength of 1400 MPa or greater. The thickness of this steel plate should be greater than or equal to 4.5 mm and less than or equal to 25 mm.

[0039] EP 2 789 699 A1 has disclosed a high-strength, hot-rolled steel product and a method for producing it. The method includes the steps of melting a steel with the following composition: C 0.25 to 0.45%, Si 0.01 to 1.5%, Mn 0.4 to 3.0%, Ni 0.5 to 4%, Al 0.01 to 1.2%, Cr $< 2\%$, Mo $< 1\%$, Cu $< 1.5\%$, V $< 0.5\%$, Nb $< 0.2\%$, Ti $< 0.2\%$, B $< 0.01\%$, Ca $< 0.01\%$, with the remainder being comprised of iron and inevitable impurities; the steel melt is cast into a slab ingot and the slab ingot is heated to a temperature in the range from 950°C . to 1350°C ., followed by a heat compensation step; the slab ingot is then hot-rolled in a temperature range from Ac_3 to 1300°C . and then immediately cooled; the cooling temperature is below the Ms -temperature and the austenite grain structure of the steel product is elongated in the rolling direction so that the length-to-width ratio is 1.2.

[0040] US 2007/0272333 A1 has disclosed a hot-rolled product, which should have a high strength; the steel has a composition comprising 0.03 to 0.1% carbon, 0.2 to 2% silicon, 0.5 to 2.5% manganese, 0.02 to 0.1% aluminum, 0.2 to 1.5% chromium, and 0.1 to 0.5% molybdenum; with 80% by area having a martensitic structure, at least in the longitudinal direction.

[0041] EP 2 340 897 A1 has disclosed a thermomechanical processing method for heavy plates. This method serves to increase the toughness, in particular the low-temperature toughness. For the production, the heavy plate is heated, partially and completely formed by means of rolling, and then is subjected to an accelerated cooling as compared to a cooling at ambient temperature; the heavy plate, which has been heated to a temperature above the Ac_3 temperature for a partial forming, is subjected to an accelerated cooling after its final forming. In order to achieve exceptional toughness values, between the partial forming and the final forming, the heavy plate is subjected to an accelerated cooling to a temperature below the ar_3 temperature and then is inductively heated to a temperature above the Ac_3 temperature.

[0042] CA 2 845 471 has disclosed a coiled steel tube, which is produced from a plurality of welded strips, wherein the tube comprises base metal regions, weld joints, and heat affected zones, and has a tensile strength of greater than 80 ksi; in addition to iron, it can contain 0.17 to 0.35 wt % carbon, 0.3 to 2 wt % manganese, 0.1 to 0.3 wt % silicon, 0.01 to 0.04 wt % aluminum, up to 0.01 wt % sulfur, and up to 0.015 wt % phosphorus, and the microstructure comprises more than 90% by volume of tempered martensite, wherein the microstructure should be homogeneous across all regions, namely the base metal regions, weld joints, and heat affected zones, and wherein the microstructure should comprise a uniform distribution of carbides. In addition, the composition can comprise up to 1 wt % chromium, 0.5 wt % molybdenum, 0.003 wt % boron, up to 0.03 wt % titanium, up to 0.5 wt % copper, up to 0.5 wt % nickel, up to 0.1 wt % niobium, up to 0.15 wt % vanadium, and up to 0.05 wt % calcium, with a maximum oxygen content of up to 0.0050 wt %.

[0043] JP 2006 183139 A has disclosed an automobile part, which is made out of a steel alloy that contains fine carbides. The fine precipitation of V-containing carbides in steel produces a particle size of 20 nm or less. The steel alloy contains C: 0.10 to 0.25%, Si: 1.5% or less, Mn: 1.0 to 3.0%, P: 0.10% or less, S: 0.005% or less, Al: 0.01 to 0.5%, N: 0.010% or less. It contains 0.10 to 1.0% V and satisfies the formula $(10\text{Mn}+\text{V})/\text{C}\geq 50$. The rest consists of Fe and inevitable impurities, with the volumetric fraction of the tempered martensite phase amounting to 80% or more.

[0044] US 2015 01017712 A1 has disclosed a high-strength cold rolled steel; the steel is kept for at least 50 s at a temperature at or above the ac_3 point, then it is cooled to an arbitrary temperature between 300° C. and 400° C. at an average cooling speed of at least 15 K/s, is kept for 5 to 180 s in a temperature range between 400° C. and 540° C., is kept in this temperature range for at least 50 s, and is then cooled. The steel has a tensile strength of at least 980 MPa.

[0045] JP 2018 021233 A has disclosed a high-strength steel sheet in which the steel has the following alloy composition: C 0.15% to 0.35%, Si+Al 0.5% to 3.0%, Mn 1.0% to 4.0%, P 0.05% or less and S 0.01% or less, with the rest made up of Fe and inevitable impurities. The steel structure contains a ferrite fraction of 5% or less, the total fraction of tempered martensite and tempered bainite is 60% or more, and the quantity of residual austenite amounts to 10% or more. MA has an average size of 1.0 μm or less. The quantity of residual austenite is 0.3% or more.

[0046] EP 2239343 A1 has disclosed a method for producing a hollow body. In particular, an electric resistance-welded steel pipe, which is made of a steel sheet and in which the width of a low-carbon layer is [sic], is subjected to a heat treatment that includes a quenching treatment. In the quenching treatment, the electric resistance-welded steel pipe is heated to a heating temperature, which is not lower than the Ac_3 transformation temperature, is kept there for a soaking time, immediately cooled to a quenching start temperature at a primary cooling rate, and then secondarily cooled (quenched). The quenching start temperature is higher than the Ar_3 transformation temperature. This prevents the hardness from being reduced during the quenching of the electric resistance-welded steel part and increases the durability of the heat treated article.

[0047] “Investigation of the hardness-toughness relationship” by Th. Schlagerl et al. is an investigation into the relationship between hardness and toughness with various forms of heat treatment. A steel wire made of T 69 5 ZMn2NiCrMo M M1 H5 is used for the investigation. The influence of the cooling time and the holding time on the ratio of hardness to toughness is investigated. The cooling time from 1350° C. to 20° C. and the holding time with an annealing temperature of 580° C. are compared. A slow cooling within 20 s promotes toughness whereas a long holding time reduces the toughness.

[0048] “Ultra High Strength Steels Produced by Thermomechanical Hot Rolling-Advanced Properties and Applications” by M. Klein et al. discloses an advantageous combination of thermomechanical rolling and subsequent martensitic transformation. It is described as a suitable method for obtaining ultra-high-strength hot rolled material with a balanced ductility and toughness. With this method for producing strips, it is possible to achieve yield strengths of 900 MPa to 1100 MPa with very low carbon concentrations of between 0.08 and 0.17 m %.

[0049] All in all, the prior art has disclosed hot rolled ultra-high-strength or wear-resistant steels for all possible forms of use, which have a high strength accompanied by a high toughness and a good processability. In this connection, products such as wide strip sheets and slab products are supplied; in particular, these are produced in wide strip mills. The rolling processes used include conventional or normalizing hot rolling (HR) and thermomechanical rolling (TM). Hot strips of this kind, produced with the conventional hot rolling methods or with the thermomechanical rolling method, are produced either by slow cooling or quenching and direct hardening (DQ) after rolling.

[0050] Pipes or profiles can also be produced using the rolling method; this is done using either seamless pipe-rolling mills or so-called roll profiling mills. The forming methods used in this connection are conventional hot rolling, thermomechanical rolling, and roll profiling. Pipes of this kind also involve a subsequent heat treatment; this heat treatment is a conventional hardening, i.e. a

pipe hardening, a conventional quenching and tempering, i.e. a pipe quenching and tempering and local weld seam finishing treatment after welding procedures;

[0051] it is not unknown to use inductive heat treatments for normalizing the hardening and the quenching and tempering.

[0052] Strips, sheets, and slab products also involve performing a subsequent heat treatment; this, too, is either a conventional hardening, e.g. slab hardening, or a conventional quenching and tempering, e.g. slab quenching and tempering; the annealing can also be performed as a slab annealing or bell annealing. Here, too, a wide variety of welding processes are performed; local weld seam finishing treatments are customary.

[0053] In the previous methods for heat treatment of such steel grades and steel products, problems arise. Basically, conventional hardening or quenching and tempering can only be performed with piece goods. These are sheets that have been cut to size or pipes or profiles that have been cut to length. Basically, this is quite laborious and therefore also cost-intensive. Such conventionally hardened products frequently have higher alloying concentrations, in particular C concentrations, which have a negative effect on the weldability.

[0054] Furthermore, it is a known problem that welded products have non-homogeneous properties in the vicinity of the weld seam due to the heat affected zones.

SUMMARY OF THE INVENTION

[0055] Hot strip products as defined by the application are assumed to usually have a sheet thickness of 1.5 to 20 mm, in particular 3 to 15 mm.

[0056] The object of the invention is to establish a method for producing conventionally rolled and profiled hot strip products, which in comparison to conventionally produced hot strip products, have outstanding strength and toughness combinations and a fine isotropic structure.

[0057] The object is attained with a method for producing conventionally rolled and profiled hot strip products with the features described and claimed herein.

[0058] The object is attained with a method for producing profiled hot strip products that are rolled in a normalizing fashion, with the features described and claimed herein.

[0059] Advantageous modifications are also described and claimed herein.

[0060] All percentage indications throughout the following description are expressed in percentage by weight unless otherwise indicated.

[0061] In conventional hot-rolling, the greatest part of the forming takes place above the recrystallization stop temperature, as a result of which the austenite develops a globular grain form, as shown in FIG. 1.

[0062] The hot strip product according to the invention has a predominantly martensitic structure, which is generated from globular, fine austenite grains and therefore has homogeneous isotropic properties. This also applies to weld seams that are present.

[0063] According to the invention, however, the heat treatment is performed differently from the state of the art as a short-term heat treatment. In this connection, the short-term heat treatment according to the invention can be an inductive hardening or an inductive quenching and tempering (hardening and annealing). The short-term heat treatment, however, can be carried out with all forms of heating that enable a short-term, preferably rapid heating;

[0064] hardening is performed at least once and the annealing is optional. For it, a globular, fine austenite grain is achieved, which after its transformation into predominantly martensitic structure has maximum strength and toughness values.

[0065] According to the invention, a “short-term heat treatment” is understood, for example, to mean a hardening, which is performed once or multiple times; the heating rates are up to 1000 K/s depending on the cross-section of the product to be heated; this heating rate can decrease with increasing cross-section. The maximum temperature in this case is above A_{c3} , in other words, 800° C. to 1000° C., in particular 820° C. to 970° C. The holding time for which the maximum temperature is maintained is 0.5 to 60 seconds; finally, a cooling is performed in which the cooling

rates are between 10 K/s and up to greater than 60 K/s.

[0066] An optional annealing is performed at temperatures below A_c , with the temperatures particularly lying between 300° C. and 700° C.

[0067] To improve the weld seam properties, an annealing temperature of between 500° C. and 700° C. can be advantageous, but in order to increase the yield strength, a lower annealing temperature of 300° C. to 450° C. can be particularly advantageous.

[0068] For the method according to the invention, it is particularly suitable to use a steel that has the following composition (all values in wt %): [0069] 0.03 to 0.22% carbon, [0070] 0.0 to 2.0% silicon, [0071] 0.5 to 3.0% manganese, [0072] 0.02 to 1.2% aluminum, [0073] 0 to 2.0% chromium, [0074] 0 to 2.0% nickel, [0075] 0.0 to 1.0% molybdenum, [0076] 0.0 to 1.5% copper, [0077] 0 to 0.02% phosphorus, [0078] 0 to 0.01% sulfur, [0079] 0 to 0.008% nitrogen, [0080] 0 to 0.005% boron, [0081] 0.0 to 0.2% niobium, [0082] 0.0 to 0.3% titanium, [0083] 0.0 to 0.5% vanadium

the remainder being comprised of iron and smelting-related impurities,

[0084] The following alloy composition is particularly suitable (all values in wt %): [0085] 0.055 to 0.195 carbon, [0086] 0.0 to 0.3% silicon, [0087] 1.4 to 2.3% manganese, [0088] 0.02 to 0.6% aluminum, [0089] 0 to 2% chromium, 0 to 2% nickel, [0090] 0.0 to 0.42% molybdenum, [0091] 0.0 to 0.5% copper, [0092] 0 to 0.008% phosphorus, [0093] 0 to 0.0015% sulfur, [0094] 0 to 0.007% nitrogen [0095] 0 to 0.005% boron, [0096] 0.0 to 0.2% niobium, [0097] 0.0 to 0.3% titanium, [0098] 0.0 to 0.5% vanadium

the remainder being comprised of iron and smelting-related impurities,

[0099] With the invention, it is advantageous that it is possible to produce ultra-high-strength profiled hot strip products with significantly improved properties with regard to toughness and isotropy; a good processability and in particular, a good weldability are present and in this case, it is possible to replace conventionally quenched and tempered profiles. This particularly concerns closed, welded profiles and open profiles; an additional advantage is that it is possible to eliminate a component hardening or component quenching and tempering and that components of this kind can also be subjected to inline heat treatment by means of ultrafast heating.

[0100] In the invention, the term “inline” is understood to mean that the entire heat treatment procedure can be performed continuously even before the finished profiles are cut to length and it is advantageously possible to eliminate a separate manipulation of individual profiles.

[0101] With tubes, relatively inexpensive-to-produce welded tubes can be used as a substitute for seamless structural tubing, which results in a lower production cost and makes it possible to ensure tighter thickness tolerances.

[0102] The advantages become particularly clear when the conventional heat treatment is compared to the new short-term heat treatment.

[0103] In conventional hardening, the steel products are heated to above $A_{sub.c3}$, e.g. 920° C., and are kept there for several minutes (e.g. 10 minutes) and are then subjected to accelerated cooling. In conventional quenching and tempering, after the hardening step, an annealing treatment is performed; the temperature is below $A_{sub.c1}$, e.g. 570° C., and the annealing times are several minutes long (e.g. 15 minutes).

[0104] In the short-term heat treatment according to the invention, the hardening takes place e.g. at 950° C., but there is only, a one-second holding time for example, whereas in the quenching and tempering, the first heat treatment takes place at for example 950° C. for one second, for example, and the quenching and tempering step takes place at 650° C., for example, likewise for one second, for example.

[0105] Since for the mechanical properties, the heating rate on the one hand, but also the duration of the heat treatment particularly above the A_{c3} point can exert an influence and can also be interchanged with each other in a predictable way (more time, lower temperature and vice versa) the Hollomon-Jaffee parameter (HJP) was developed, which maps the two influence variables. The

applicant subsequently developed this further in order to also be able to provide meaningful results for continuous heat treatment processes i.e. for the heating, the holding at a maximum temperature, and the cooling (Hubmer G., Ernst W., Klein M., Sonnleitner M., Spindler H.: A TRIBUTE TO HOLLOMON & JAFFE—THE 70TH BIRTHDAY OF A BRILLIANT EQUATION, Proc. 6th Int. Conf. on Modeling and Simulation of Metallurgical Processes in Steelmaking (STEELSIM 2015), Bardolino (2015)).

[0106] Particularly advantageous mechanical properties, especially for the product of notched bar impact work KV and tensile strength Rm, can result if the HJ Parameter of the hardening process is set to between 18000 and 23000, preferably between 18500 and 22000 with a heating to a maximum temperature of 800° C. to 1000° C., in particular 820° C. to 970° C.

[0107] With the method according to the invention, profiles and tubes can be produced, which have a particularly good combination of a high tensile strength Rm and high notched bar impact bending work KV, particularly at low temperatures. The product of Rm*KV can be >45,000 MPaJ, preferably >50,000 MPaJ, particularly preferably >60,000 MPaJ, and especially >65,000 MPaJ.

[0108] In general, it should be noted that the notched bar impact work KV was measured at -40° C.; it is to be expected that the value would have turned out to be even higher at a higher temperature.

[0109] The invention thus relates to a method for producing conventionally hot-rolled hot strip products; a steel alloy is melted; the melted steel alloy is cast into slab ingots and after being heated to a temperature above Ac.sub.3, the slab ingots are hot rolled until they reach a desired degree of deformation and a desired strip thickness; the rolling is performed above the recrystallization temperature of the alloy; after the rolling, the strip is cooled to room temperature and for hardening purposes, is briefly heated to a temperature >Ac3 and cooled again, characterized in that the heating takes place with a temperature increase of more than 5 K/s, preferably with more than 10 K/s, particularly preferably with more than 50 K/s, especially with more than 100 K/s and is kept at a desired target temperature for 0.5 to 60 s and then a cooling takes place; the steel strip or blank, which has been produced by means of normalizing rolling, is profiled into a component in a forming process.

[0110] In a modification, a steel alloy is melted, which contains the following elements as well as iron and inevitable impurities, each expressed in wt % [0111] 0.03 to 0.22% carbon, [0112] 0.0 to 2.0% silicon, [0113] 0.5 to 3.0% manganese, [0114] 0.02 to 1.2% aluminum, [0115] 0 to 2.0% chromium, [0116] 0 to 2.0% nickel, [0117] 0.0 to 1.0% molybdenum, [0118] 0.0 to 1.5% copper, [0119] 0 to 0.02% phosphorus, [0120] 0 to 0.01% sulfur, [0121] 0 to 0.008% nitrogen, [0122] 0 to 0.005% boron, [0123] 0.0 to 0.2% niobium, [0124] 0.0 to 0.3% titanium, [0125] 0.0 to 0.5% vanadium

the remainder being comprised of iron and smelting-related impurities

[0126] In another advantageous embodiment, a steel alloy is melted, which particularly contains the following elements as well as iron and inevitable impurities, each expressed in wt % [0127] 0.055 to 0.195 carbon, [0128] 0.0 to 0.3% silicon, [0129] 1.4 to 2.3% manganese, [0130] 0.02 to 0.6% aluminum, [0131] 0 to 2% chromium, 0 to 2% nickel, [0132] 0.0 to 0.42% molybdenum, [0133] 0.0 to 0.5% copper, [0134] 0 to 0.008% phosphorus, [0135] 0 to 0.0015% sulfur, [0136] 0 to 0.007% nitrogen [0137] 0 to 0.005% boron, [0138] 0.0 to 0.2% niobium, [0139] 0.0 to 0.3% titanium, [0140] 0.0 to 0.5% vanadium

the remainder being comprised of iron and smelting-related impurities

[0141] Advantageously, the brief heating can be carried out with all suitable forms of heating, e.g. inductive.

[0142] In addition, the target temperature in the brief heating for hardening purposes can advantageously be >Ac.sub.3, which means 800° C. to 1000° C., in particular 820° C. to 970° C.

[0143] In another advantageous modification, the target temperature in the brief heating for annealing purposes is <Ac.sub.1, with the temperatures in particular being between 300° C. and

700° C.

[0144] In another advantageous embodiment, the holding times at the target temperature in the hardening and/or annealing and/or quenching and tempering are 0.5 to 10 s, in particular less than 5 seconds.

[0145] The cooling after the heating step or steps can also advantageously take place at cooling rates of $>10^{\circ}$ K/s.

[0146] Even more preferably, the cooling rate can advantageously be >30 K/s and in particular >60 K/s.

[0147] In an alternative embodiment, direct hardening can be performed from the rolling heat (DQ).

[0148] In an advantageous embodiment, the steel strip can be subjected to the short-term heat treatment before the profiling or the profiled component can be subjected to the short-term heat treatment.

[0149] It can be advantageous if after the forming and welding, a short-term heat treatment is performed to homogenize the weld seam.

[0150] In a particularly advantageous embodiment, the sheet thickness can be 1.5 mm to 20 mm, in particular 3 mm to 15 mm.

[0151] With the invention, it is advantageous if the Hollomon-Jaffee parameter of the short-term hardening process is between 18000 and 23000, preferably between 18500 and 22000.

[0152] The invention also relates to a profiled component produced with one of the abovementioned methods, in which at least one of the following mechanical properties is satisfied

[0153] tensile strength (R_m) ≥ 1200 MPa [0154] notched bar impact bending work (KV) ≥ 40 J

[0155] and the following condition is satisfied [0156] $R_m \times KV \geq 50000$ MPa J

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0157] The invention will be explained by way of example based on the drawings. In the drawings:

[0158] FIG. 1 shows the influence of conventional hot rolling on the structure;

[0159] FIG. 2 shows the influence of the thermomechanical rolling on the structure;

[0160] FIG. 3 shows the difference in the microstructure between recrystallized austenite and non-recrystallized austenite;

[0161] FIG. 4 shows the steel phases based on the temperature curves produced;

[0162] FIG. 5 shows the comparison of heat treatment routes in conventionally hot-rolled and conventionally quenched and tempered products and a heat-treated product according to the invention;

[0163] FIGS. 6a/6b show the temperature/time curves for the treatment routes in FIG. 5 that are not according to the invention and the structures that are finally established;

[0164] FIG. 7 shows the product of the tensile strength R_m and the notched bar impact work KV as a function of the Hollomon-Jaffee parameter of the hardening process for short-term hardening procedures according to the invention and for conventional hardening of the steel that has been heat treated according to the invention (material C described in Table 1) in contrast to conventionally heat treated steels;

[0165] FIGS. 8a/8b show the possible temperature/time curves in the method according to the invention with the structure that is established in the individual production steps.

DETAILED DESCRIPTION OF THE INVENTION

[0166] According to the invention, steel is conventionally hot-rolled and subjected to a short-term heat treatment in order to increase the properties of toughness and isotropy as well as other properties.

[0167] According to FIG. 1, conventionally hot rolled steels, steels in which the rolled product is first heated to the hot-forming temperature and then rolled, by means of which the non-deformed grain is deflected in the rolling direction; already during the rolling, a recrystallization takes place after each roll pass, at the end of which the respective austenite grain has a globular form.

[0168] By contrast with this, thermomechanically rolled steels contain higher concentrations of carbide-forming elements, which form precipitation already during the hot rolling. The precipitation and the dissolved micro-alloying elements delay or suppress the recrystallization after the roll passes. Correspondingly, a recrystallization and a corresponding grain growth do not occur so that according to FIG. 2, a globular structure according to FIG. 1 is not formed and instead, the austenite is in an elongated form.

[0169] In FIG. 3 the different austenite embodiments are shown, on the one hand, the globular recrystallized austenite (top) and on the other, the elongated, non-recrystallized austenite (bottom).

[0170] The difference between the conventionally hot-rolled steels with the globular recrystallized austenite grain on the one hand and the thermomechanically rolled steels with the non-globular, elongated, and deformed austenite grain is that the austenite grain of the thermomechanically rolled steel exhibits a different structure after the transformation.

[0171] Correspondingly, the forming has significant effects on the structure and properties; the properties cannot be achieved by means of the heat treatment alone.

[0172] FIG. 4 schematically depicts how from the austenite range, by means of different cooling curves, it is also possible to achieve different structures or microstructures. It shows that by means of different cooling paths, martensitic steels, complex-phase steels, dual-phase steels, and ferritic-bainitic steels can be achieved.

[0173] Prior conventional heat treatment routes are shown in FIG. 5, lines 1 and 2. For example, the hot-rolling and a conventional quenching and tempering step (a slab quenching and tempering), which is used for sheets, and the conventional hot-rolling, which can be combined with a direct hardening step (DQ) and an annealing step (A).

[0174] The method according to the invention (FIG. 5, last line) provides a normalizing conventional hot-rolling and an optional direct hardening (with an optional annealing step), and then at least one very short-term, for example inductive, hardening step or quenching and tempering step.

[0175] The temperature/time curves according to the prior art are shown in FIGS. 6a and 6b.

[0176] Before this short-term inductive hardening step or quenching and tempering step, the hot strip is allowed to cool or is cooled to room temperature (e.g. after the direct hardening). A further processing from the rolling heat does not take place.

[0177] In comparison to the conventionally rolled, direct-hardened, and annealed processing route (FIG. 5, middle line), according to the invention, a conventional rolling, direct hardening, and at least one very short-term-for example inductive-quenching and tempering step are performed.

[0178] The differences in the structures are clear when known structures shown in FIG. 6a and FIG. 6b are compared to the structure produced according to the invention shown in FIG. 9a. The structure of the hot-rolled and short-term heat-treated steel according to the invention differs significantly from that of the conventionally treated steel; the smaller size and more isotropic form of the grain structure are particularly conspicuous.

[0179] Basically, the quenching and tempering step should be explained once again; the conventional quenching and tempering step is shown in FIG. 6a.

[0180] In the conventional quenching and tempering, a product is first heated in a reheating furnace and is then conventionally hot-rolled in a normalizing way and completely cooled.

[0181] For the quenching and tempering, it is heated again to approx. 900° C. and then a rapid cooling in water is performed, followed by an annealing step at approx. 600° C. with a subsequent cooling in air.

[0182] The conventional heat treatments that are not according to the invention are thus the

conventional hardening (H) or slab hardening, the conventional quenching and tempering (H+A) or slab quenching and tempering, and the conventional annealing (A) in the form of slab annealing or bell annealing.

[0183] In the conventional hardening or quenching and tempering, it is only possible to treat piece goods, which is relatively costly. In conventional thermomechanical rolling, the elongation of the structure produces an anisotropy of the properties; a slab annealing can achieve very good strength/toughness ratios, but it is only possible to heat treat slabs and not strips.

[0184] By contrast with conventional methods, according to the invention, the subsequent heat treatments (H.sub.ST, A.sub.ST) are performed as short-term heat treatments.

[0185] By contrast with the prior art, in the heating according to the invention, as shown in FIG. 8a, a rapid short-term heating is performed; for example, the heat source can be an inductive heating, but does not have to be.

[0186] According to the invention, hardening can be performed at least once and annealing can optionally be performed once. This yields a globular, fine austenite grain with a maximized strength and a maximized toughness.

[0187] According to the invention, the hardening can be performed once or twice; at 100 to 1000° K/s, the heating rates can be very high; the maximum temperature is set to >AC3. According to the invention, this temperature is 800° C. to 1000° C., in particular between 820° C. and 970° C. The holding time is extremely short compared to the prior art and can be from 0.5 to 60 seconds, in particular from 0.5 to 5 seconds.

[0188] According to the invention, however, the heating rate can also be selected as lower and can, for example, be 5 K/s, 10 K/s, or 15 K/s.

[0189] Essential to the invention, however, are the short holding times of 0.5 to 60 seconds, preferably 0.5 to 20 seconds, in particular 0.5 to 5 seconds.

[0190] The subsequent cooling rates are set anywhere from >10° K/s up to greater than 60° K/s.

[0191] The optional annealing is performed at a maximum temperature below Ac, which is normally from 300° C. to 700° C. In order to avoid a softening zone in subsequent welding processes, an annealing temperature of between 500° C. and 700° C. can be advantageous, but in order to increase the yield strength, a lower annealing temperature of 300° C. to 450° C. can be particularly advantageous.

[0192] The short-term heat treatments according to the invention are thus one the one hand hardening treatments or quenching and tempering treatments.

[0193] Table 1 shows select properties of a steel (Material C) that has been heat treated according to the invention in contrast to conventionally heat treated steels. FIG. 7 shows examples of the properties that can be achieved as a function of the heat treatment routes and parameters for the alloy composition.

TABLE-US-00001
TABLE 1 Properties of Material C, Heat Treated According To Invention
Material C Si Mn P S Al Cr Ni Mo Cu V Nb Ti B N Material C 0.172 0.18 2.29 0.008 0.0006 0.051 0.27 0.02 0.024 0.08 0.005 0.002 0.019 0.0022 0.0034 R.sub.p.02 R.sub.m KV@-40° C. Rm .Math. KV@-40° C. Material Production process [MPa] [MPa] [J] [MPa .Math. J] Material C Prior art HR + H 1076 1539 26 40,014 H: 920° C., 10', HJ = 23,380 HR + DQ 1333 1577 20 31,540 Invention HR + DQ + (A) + H.sub.ST 998 1410 48 67,680 H.sub.ST: 950° C., 1'', HJ = 21,882

[0194] In Table 1, the material indicated in the table (material C) is on the one hand subjected to a heat treatment with two different routes according to the prior art; first, after the hot-rolling (HR), it is kept at 920° C. for 10 minutes for a hardening process. The HJ value in this case is 23380. This yields the mechanical properties RP02 of MPa, Rm of 1539 MPa, and the relatively low notched bar impact bending work of 26 J. The product of Rm and KV is about 40000 MPaJ. Alternatively, a direct hardening step (DQ) can also be carried out, but this does not significantly improve the mechanical properties.

[0195] If, however, as already described above, the material C is treated using the method

according to the invention with a short-term heat treatment (HST), with a temperature of 950° C. being maintained for 1 second in the example, then the short-term heat treatment can increase the tensile strength over that of a conventionally produced, hot-rolled product, but in particular, the toughness is improved to a quite considerable degree. Optionally, the HST can also be preceded by an annealing step, but this was not performed in this exemplary embodiment. The HJ parameter is 21882. In this example, mechanical properties Rp02 of 998 MPa and Rm of 1410 MPa, but above all, an outstanding notched bar impact bending work of 48 J are achieved. The product of Rm and KV in this case is 67,680 MPaJ.

[0196] In FIG. 7, the product of tensile strength and notch bar impact work at -40° C. as a function of the HJ parameter is plotted for different hardening processes. The light point corresponds to the above-described example A according to the invention with an HJ of 21,882 and the dark point corresponds to the prior art. The HJ value should be between 18000 and 22000 and the maximum temperatures should be in the range of 800-1000° C. With an excessively low HJP and excessively low maximum temperatures, a complete austenitization does not occur and the material cannot be completely hardened. The HJP and the maximum temperature of the hardening process, however, must not be selected too high and in particular, the HJP must be below 23000 since otherwise, the mechanical properties (especially the product of Rm and KV) can decrease drastically.

[0197] FIG. 8a shows the temperature/time curve according to a possible embodiment of the invention together with the structures that are established.

[0198] By means of the processing step or production step of welding, the introduced energy (heat and/or pressure) causes a local change in the structure and the mechanical properties. Products therefore have nonhomogeneous properties in the region of the weld seam.

[0199] If in the course of production, the short-term heat treatment according to the invention is used after a processing step of “welding,” then as shown in FIG. 8b for a fusion welding process, a homogenization of the microstructure occurs in the weld seam region. The microstructure of the weld seam region and also its mechanical properties are thus brought into line with those of the rest of the product.

[0200] This is true for both fusion-welded connections such as laser welds and for pressure-welded connections such as high-frequency welds.

[0201] The invention will be explained in greater detail based on an example:

[0202] The product according to the invention is produced in that first, a steel melt with the composition according to the invention, particularly the chemical composition indicated in Table 1, FIG. 7 or 8, is melted in the steel mill and after the secondary metallurgical treatment, is cast into a slab ingot in a continuous casting machine.

[0203] The slab ingot is then heated to a temperature in the range from 1100° C. to 1300° C., in particular 1200° C. to 1260° C., descaled, and then conventionally hot rolled into a steel strip; in the hot rolling of the slab ingot, the initial rolling temperature is in the range from 1000° C. to 1250° C. and the final rolling temperature is greater than 800° C. and in particular, is between 830° C. and 930° C. In this case, the greatest part of the forming takes place above the recrystallization stop temperature as a result of which, the austenite develops a globular grain form, as shown in FIG. 1. After the hot rolling, the steel strip is cooled from the final rolling temperature to the coiling temperature by means of water exposure and is coiled. In the present example, the coiling temperature is below the martensite start temperature, i.e. less than 500° C., in particular less than 250° C., and is achieved at a cooling rate of greater than 25° C./s, in particular between 40° C./s and 100° C./s.

[0204] The steel strip, with or without a preceding blank cutting (e.g. cross-cutting or longitudinal cutting), is optionally subjected to a heat treatment; the temperature assumes values below the A1 temperature, in particular below 700° C.

[0205] Blanks made of the steel strip produced according to the invention can optionally be connected by means of a welding process. In this case, these blanks can have different dimensions

or chemical compositions.

[0206] The steel strip or blank is formed into a component (profile, tube, edge piece) in a forming process with an optionally integrated welding process (e.g. in a roll-profiling machine or bending machine).

[0207] According to the invention, the steel strip is subjected to a short-term heat treatment before the forming or else the formed component (profile, tube, edge piece) is subjected to a short-term heat treatment.

[0208] In this case, the product is first heated at least once to a maximum temperature above A_{c3} ; typically, this is 800°C . to 1000°C ., in particular however 820°C . to 970°C ., briefly kept at this temperature, and then rapidly cooled. The heating rates, depending on the cross-section of the product to be heated, are greater than 5 K/s, preferably greater than 10 K/s, particularly preferably greater than 50 K/s, in particular greater than 100 K/s. The holding time at the maximum temperature is 0.5 to 60 seconds, for example 1 to 10 s; then, a cooling is performed at cooling rates between 10 K/s and up to greater than 60 K/s.

[0209] After the hardening, the steel strip can be subjected to another annealing treatment before the forming or else the formed component (profile, tube, edge piece) can be subjected to another annealing treatment. In the latter, the sheet is heated at a heating rate of up to 1000 K/s, in particular $400\text{--}800^{\circ}\text{C}/\text{s}$, to a maximum temperature below A_{c1} , which usually means 300°C . to 700°C ., for example 550°C . The holding time at the maximum temperature is 0.5 to 60 seconds, for example 1-10 s; then, a cooling is performed at cooling rates between 10 K/s and up to greater than 60 K/s.

[0210] The invention will be explained in greater detail based on a specific example:

[0211] The product according to the invention is produced in that first, a steel melt with the composition according to the invention, in particular the chemical composition indicated in FIG. 7 [sic], is melted in the steel mill and after secondary metallurgical treatment, is cast into a slab ingot in a continuous casting machine.

[0212] The slab ingot is then heated to a temperature of 1245°C ., descaled, and then conventionally hot-rolled into a steel strip; in the hot rolling of the slab ingot, the initial rolling temperature is 1140°C . and the final rolling temperature is 865°C . In this case, the greatest part of the forming takes place above the recrystallization stop temperature as a result of which, the austenite develops a globular grain shape, as shown in FIG. 1. After the hot rolling, the steel strip is cooled from the final rolling temperature to the coiling temperature by means of water exposure and is coiled.

[0213] In the present example, the coiling temperature is 125°C . and is achieved at a cooling rate of $55^{\circ}\text{C}/\text{s}$.

[0214] The steel strip, which has been cut to length and has a thickness of 5 mm, is formed into a closed profile with a square cross-section and the outer dimensions of 50 mm×50 mm in a roll-profiling machine with an integrated welding process.

[0215] According to the invention, the profile is subjected to a short-term heat treatment.

[0216] In this case, the profile is initially heated to a maximum temperature above A_{c3} , to 950°C . in the present example, is briefly held at this temperature, and is then rapidly cooled. The heating rates are 15 K/s. The holding time at the maximum temperature is 1 second; finally, a cooling at cooling rates of 20 K/s is performed. The Hollomon-Jaffee parameter of the short-term hardening that is performed is 21882.

Claims

1. A continuous, inline method for producing hot-rolled hot strip products having a predominantly martensitic structure generated from globular, fine austenitic grains, comprising the steps of: providing a steel alloy including the following elements, in percent by weight: 0.03 to 0.22%

carbon, 0.0 to 2.0% silicon, 0.5 to 3.0% manganese, 0.02 to 1.2% aluminum, 0 to 2.0% chromium, 0 to 2.0% nickel, 0.0 to 1.0% molybdenum, 0.0 to 1.5% copper, 0 to 0.02% phosphorus, 0 to 0.01% sulfur, 0 to 0.008% nitrogen, 0 to 0.005% boron, 0.0 to 0.2% niobium, 0.0 to 0.3% titanium, 0.0 to 0.5% vanadium, the remainder being comprised of iron and smelting-related impurities melting the steel alloy; casting the melted steel alloy into slab ingots; heating the slab ingots to a temperature above A_{c3} ; hot rolling the slab ingots to produce steel strips having a desired degree of deformation and a desired strip thickness, the rolling being performed above a recrystallization temperature of the alloy; cooling the steel strips to room temperature; profiling the steel strips into one or more components using a forming process; hardening the one or more components by heating the one or more components to a temperature $>A_{c3}$ and cooling the one or more components again to form hardened one or more components; wherein the heating of the one or more components takes place with a temperature increase of more than 5 K/s, and the one or more components are kept at a desired target temperature for a holding period of 0.5 to 60 s prior to cooling and an optional annealing.

2. The method according to claim 1, wherein the steel alloy comprises the following components in percent by weight: 0.055 to 0.19% carbon, 0.0 to 0.3% silicon, 1.4 to 2.3% manganese, 0.02 to 0.6% aluminum, 0 to 2% chromium, 0 to 2% nickel, 0.0 to 0.42% molybdenum, 0.0 to 0.5% copper, 0 to 0.008% phosphorus, 0 to 0.0015% sulfur, 0 to 0.007% nitrogen 0 to 0.005% boron, 0.0 to 0.2% niobium, 0.0 to 0.3% titanium, 0.0 to 0.5% vanadium, the remainder being comprised of iron and smelting-related impurities.

3. The method according to claim 1, wherein the heating of the one or more components comprises inductive heating.

4. The method according to claim 1, wherein the heating of the one or more components to a temperature $>A_{c3}$ comprises heating the one or more components to between about 800° C. and about 1000° C.

5. The method according to claim 1, further comprising the step of annealing the hardened one or more components at a temperature of about 300° C. to about 700° C.

6. The method according to claim 1, wherein the holding period is about 0.5 to about 10 seconds.

7. The method according to claim 1, wherein the step of cooling the one or more components after the heating step takes place at a cooling rate of $>30^{\circ}$ K/s.

8. The method according to claim 1, further comprising the steps of welding the one or more components to form a weld seam and heat treating the welded one or more components to homogenize the weld seam.

9. The method according to claim 1, wherein the hardened one or more components have a thickness of about 1.5 mm to about 20 mm.

10. The method according to claim 1, wherein the step of hardening the one or more components is performed using a Hollomon-Jaffee parameter of about 18000 to about 23000.

11. A profiled component produced with a method according to claim 1, wherein the profiled component comprises at least one of the following mechanical properties: tensile strength (R_m) ≥ 1200 MPa, notched bar impact bending work (KV) ≥ 40 J, measured at -40° C., and the following condition is satisfied: $R_m \times KV \geq 50000$ MPa J.

12. A use of the profiled component according to claim 11 for producing at least one of support structures in steel construction, machinery construction, automobile manufacture, and crane construction; security plates; and wear protection applications.

13. A continuous, inline method for producing hot-rolled hot strip products having a predominantly martensitic structure generated from globular, fine austenitic grains, comprising the steps of: providing a steel alloy including the following elements, in percent by weight: 0.03 to 0.22% carbon, 0.0 to 2.0% silicon, 0.5 to 3.0% manganese, 0.02 to 1.2% aluminum, 0 to 2.0% chromium, 0 to 2.0% nickel, 0.0 to 1.0% molybdenum, 0.0 to 1.5% copper, 0 to 0.04% total of phosphorus, sulfur, nitrogen and boron, 0.0 to 1.0% total of niobium, titanium and vanadium, the remainder

being comprised of iron and smelting-related impurities melting the steel alloy; casting the melted steel alloy into slab ingots; heating the slab ingots to a temperature above A_{c3} ; hot rolling the slab ingots to produce steel strips having a desired degree of deformation and a desired strip thickness, the rolling being performed above a recrystallization temperature of the alloy; cooling the steel strips to room temperature; profiling the steel strips into one or more components using a forming process; hardening the one or more components by heating the one or more components to a temperature $>A_{c3}$ and cooling the one or more components again to form hardened one or more components; wherein the heating of the one or more components takes place with a temperature increase of more than 10 K/s, and the one or more components are kept at a desired target temperature for a holding period of 0.5 to 60 s prior to cooling the one or more components again and an optional annealing.

14. The method of claim 13, wherein the heating of the one or more components takes place with a temperature increase of more than 50 K/s.

15. The method of claim 13, wherein the hot rolled strip products comprise an elongated, non-recrystallized austenite structure.

16. The method of claim 13, wherein the hot rolling comprises a plurality of roll passes and recrystallization of the steel alloy does not occur during the plurality of roll passes.

17. The method of claim 13, wherein the vanadium is present in the steel alloy in an amount of 0.0 to 0.8% by weight.

18. A continuous, inline method for producing hot-rolled hot strip products having a predominantly martensitic structure generated from globular, fine austenitic grains, comprising the steps of: providing a steel alloy including the following elements, in percent by weight: 0.03 to 0.22% carbon, 0.0 to 2.0% silicon, 0.5 to 3.0% manganese, 0.02 to 1.2% aluminum, 0 to 2.0% chromium, 0 to 2.0% nickel, 0.0 to 1.0% molybdenum, 0.0 to 1.5% copper, 0 to 0.04% total of phosphorus, sulfur, nitrogen and boron, 0.0 to 1.0% total of niobium, titanium and vanadium, the remainder being comprised of iron and smelting-related impurities melting the steel alloy; casting the melted steel alloy into slab ingots; heating the slab ingots to a temperature above A_{c3} ; hot rolling the slab ingots using one or more roll passes to produce steel strips having a desired degree of deformation and a desired strip thickness, the rolling being performed above a recrystallization temperature of the alloy; cooling the steel strips to room temperature; profiling the steel strips into one or more components using a forming process; after cooling the one or more components to room temperature, hardening the one or more components by heating the one or more components to a temperature $>A_{c3}$ and cooling the one or more components again to form hardened one or more components; wherein the heating of the one or more components takes place with a temperature increase of more than 10 K/s, and the one or more components are kept at a desired target temperature for a holding period of 0.5 to 60 s prior to cooling the one or more components again and an optional annealing; and wherein recrystallization of the steel alloy does not occur during the one or more roll passes.

19. The method of claim 18, wherein the hot rolled strip products comprise an elongated, non-recrystallized austenite structure.

20. The method of claim 18, wherein the hot rolling comprises a plurality of roll passes and recrystallization of the steel alloy does not occur during the plurality of roll passes.
